

Automated Breach & Attack Simulation Market

Size, Growth Drivers, Competitive Landscape & Strategic Opportunity

\$4.68_B

FORECAST MARKET
SIZE

RESEARCH &
MARKETS

38.4%

PROJECTED CAGR
FASTEST-GROWING
SECURITY CATEGORY

\$213_B

GLOBAL CYBER
SPEND 2025

+15.1% YOY ·
GARTNER

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Executive Summary

KEY FINDINGS

- The Automated Breach & Attack Simulation (BAS/CART) market is on a hypergrowth trajectory, projected to reach **\$4.68 billion** with a CAGR of **38.4%**, making it the fastest-growing subcategory in enterprise security (Research & Markets).
- The broader penetration testing market is expected to grow from **\$2.36 billion in 2025 to \$5.54 billion by 2031**, creating a large addressable market for automated solutions that are faster, cheaper, and continuous versus point-in-time manual engagements.
- Global cybersecurity spending reached **\$213 billion in 2025**, with 78% of enterprise executives actively planning budget increases, driven by AI-powered threats and rising breach costs.
- Venture investment in autonomous security is surging: Pentera closed a **\$60M Series D at \$1B+ valuation** (2025); Horizon3.ai (NodeZero) raised a **\$100M Series D** (June 2025) — validating investor confidence in this category.
- The competitive landscape is still fragmented. The top players — Pentera, NodeZero, Cymulate, AttackIQ, SafeBreach — each address only a portion of the attack surface. **No incumbent offers fully autonomous, AI-native, swarm-orchestrated testing** with integrated compliance reporting at sub-cent-per-scan economics.
- Chelys Security is positioned at the intersection of BAS, autonomous red-teaming, and AI-native security — the highest-growth vector within the market.

Market Overview & Size

Automated Breach and Attack Simulation (BAS) — also known as Continuous Automated Red Teaming (CART) — is a rapidly emerging class of cybersecurity technology that programmatically emulates real-world cyberattacks to identify security gaps before adversaries do. Unlike traditional penetration testing, which is manual, periodic, and expensive, BAS platforms execute continuously, safely, and at machine speed across production environments.

The category is at an inflection point. Manual penetration testing engagements average **\$25,000–\$100,000** per engagement and take 2–4 weeks to complete. Automated platforms compress this to hours at a fraction of the cost — a structural disruption similar to what cloud computing did to on-premise infrastructure.

\$1.11B

BAS MARKET SIZE (2024)

Grand View Research

\$1.48B

BAS MARKET SIZE (2025)

Market Research Future

\$4.68B

FORECAST MARKET SIZE

Research & Markets

The BAS/CART market is growing at 3–4× the rate of the overall cybersecurity market (38.4% vs. 15.1% CAGR), reflecting both the immaturity of the category and the urgency of demand. Most enterprise organizations still rely on annual manual penetration tests — the addressable market for conversion to continuous automated testing is massive.

Defining the Total Addressable Market

The TAM can be viewed from two angles:

Category TAM (BAS/CART): \$4.68B per Research & Markets at current CAGR trajectory, growing toward \$16.7B by 2035 (Precedence Research).

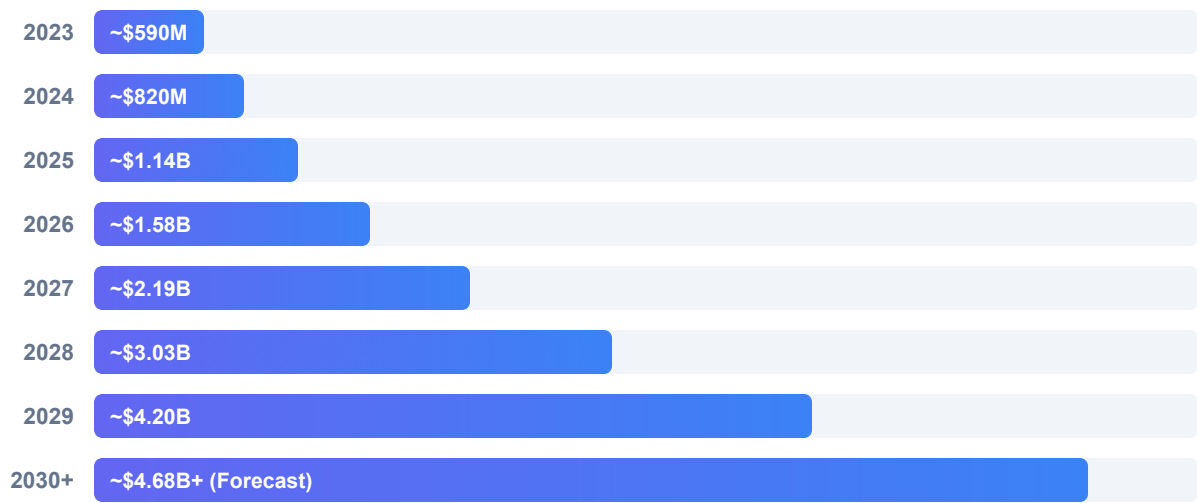
Broader TAM (Penetration Testing): The manual penetration testing market — the primary conversion opportunity — is sized at \$2.36B in 2025, growing to \$5.54B by 2031 (MarketsandMarkets). Automated platforms compete directly for this budget.

Serviceable Addressable Market (SAM) for Chelys: Mid-market and enterprise organizations with dedicated security budgets, regulatory compliance requirements, or active bug bounty programs. Estimated at 150,000+ organizations globally meeting this profile.

Growth Forecast

Multiple independent analyst firms have sized and forecast the BAS/CART market. Despite methodological differences in scope, all sources agree on the direction: exceptional compound growth driven by macro tailwinds in cyber threat intensity, AI adoption, and regulatory pressure.

BAS Market Size by Year — Illustrative Projection (38.4% CAGR)



Source: Research & Markets (\$4.68B headline); intermediate years extrapolated at published 38.4% CAGR. Independent estimates from Grand View Research, Market Research Future, and Mordor Intelligence corroborate trajectory.

Cross-Analyst Consensus

RESEARCH FIRM	2025 ESTIMATE	TERMINAL YEAR	TERMINAL VALUE	CAGR
Research & Markets	—	~2030	\$4.68B	38.4%
Market Research Future	\$1.48B	2030	\$3.0B	35.3%
Grand View Research	\$1.11B (2024)	2030	~\$3.5B	34.2%
Precedence Research	\$0.57B	2035	\$16.7B	40.1%
360iResearch	\$0.92B	2030	—	38.6%
Mordor Intelligence	~\$1.0B	2030	~\$4.0B	~35–38%

Analyst consensus: CAGR estimates across six independent research firms cluster tightly between 34% and 40%, indicating high conviction in the growth trajectory. The variance in absolute size reflects differing market definitions (BAS only vs. BAS + continuous red teaming + adversarial simulation).

Key Market Growth Drivers

The sustained high CAGR in BAS/CART is driven by converging macro trends across the threat landscape, regulatory environment, and enterprise security posture. These tailwinds are structural, not cyclical.

01

AI-Powered Threat Escalation

Nation-state actors and organized criminal groups now leverage AI to accelerate phishing, exploit development, and lateral movement. Security teams must match adversary automation with defensive automation. Gartner's 2026 Top Cybersecurity Trends explicitly calls out AI-driven attack automation as a top board-level concern, directly fueling demand for continuous automated testing.

02

Rising Cost of Data Breaches

IBM's Cost of a Data Breach Report pegged the average breach cost at **\$4.45M in 2023**, rising year-over-year. Organizations are shifting from reactive incident response to proactive continuous validation. A single avoided breach justifies 3–5 years of automated testing spend, creating a compelling ROI case for buyers.

03

Regulatory & Compliance Mandates

PCI-DSS 4.0, SOC 2 Type II, HIPAA, DORA (EU), NIS2, and CISA guidance now require documented, frequent security assessments. Annual manual penetration tests no longer satisfy regulators demanding continuous evidence of control effectiveness. BAS platforms provide the audit trail, continuous validation records, and compliance mapping regulators require.

04

Talent & Cost Scarcity in Manual Testing

There are an estimated **3.5 million unfilled cybersecurity jobs globally** (Cybersecurity Ventures). Skilled penetration testers command \$120,000–\$200,000+ in annual salary, and are in perpetual short supply. Automated platforms let organizations scale security coverage without hiring — a structural, not discretionary, pull toward automation.

05

Cloud & Hybrid Infrastructure Complexity

Cloud adoption introduces dynamic attack surfaces — ephemeral infrastructure, misconfigured IAM roles, exposed S3 buckets, container escape paths — that change daily. Static annual penetration tests cannot track this. BAS platforms provide continuous coverage of cloud-native environments, Kubernetes clusters,

06

Zero Trust Architecture Adoption

Zero Trust mandates (NIST SP 800-207, US Executive Order 14028) require organizations to continuously verify that access controls are functioning as designed — not just configured. BAS platforms are the primary mechanism for validating zero trust implementations continuously, creating a durable, policy-driven demand driver across government and regulated industries.

and multi-cloud configurations where attack surface changes constantly.

Combined effect: These six drivers are simultaneously pulling enterprises toward automated security validation. Unlike many software categories where demand is driven by a single catalyst, BAS demand has multiple independent forcing functions — making it highly resilient to individual macro shocks.

Competitive Landscape

The BAS/CART market is growing rapidly but remains fragmented. No single vendor holds dominant market share, and the top players each address different technical approaches, customer sizes, and attack coverage areas. This fragmentation represents both a validation of the category and a significant opportunity for a differentiated entrant.

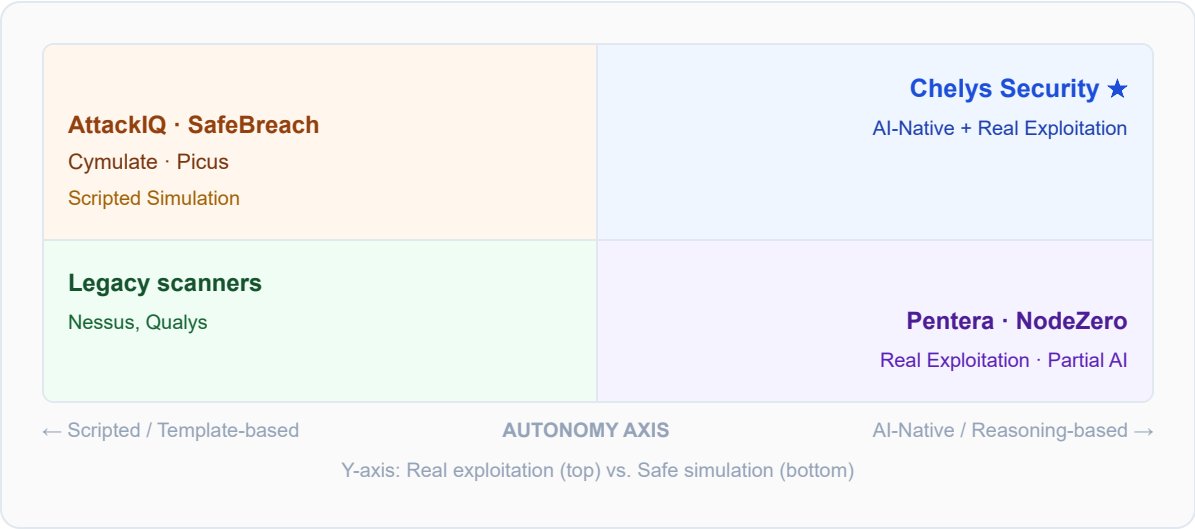
Vendor Comparison Matrix

COMPANY	FOUNDED	FUNDING	VALUATION	APPROACH	AI-NATIVE	KEY GAP
Pentera	2015	\$285M+	\$1B+	Agentless autonomous pentest	Partial	No AI reasoning; limited lateral movement chain
Horizon3.ai (NodeZero)	2019	\$186M	Unicorn est.	Autonomous internal pentest	Partial	No agent reasoning; primarily network-focused
AttackIQ	2013	\$95M	—	MITRE ATT&CK simulation library	No	Simulated attacks only; not real exploitation
SafeBreach	2014	\$106M	—	BAS playbook simulation	No	Pre-scripted; cannot adapt to novel environments
Cymulate	2016	\$141M	—	Multi-vector BAS platform	No	No real exploitation; template-driven
XM Cyber	2016	Acq. \$700M	Acquired	Attack path management	Partial	Graph simulation; not actual exploitation

COMPANY	FOUNDED	FUNDING	VALUATION	APPROACH	AI-NATIVE	KEY GAP
Picus Security	2013	~\$45M	—	BAS + threat intelligence	No	Simulation only; no exploitation chain
Chelys Security	2026	—	—	AI-native autonomous swarm pentest	Full	Entrant — see Section 9

Strategic Positioning Map

Vendors in this space can be mapped across two critical dimensions: **autonomy** (simulation vs. real exploitation) and **AI intelligence** (scripted vs. reasoning-based).

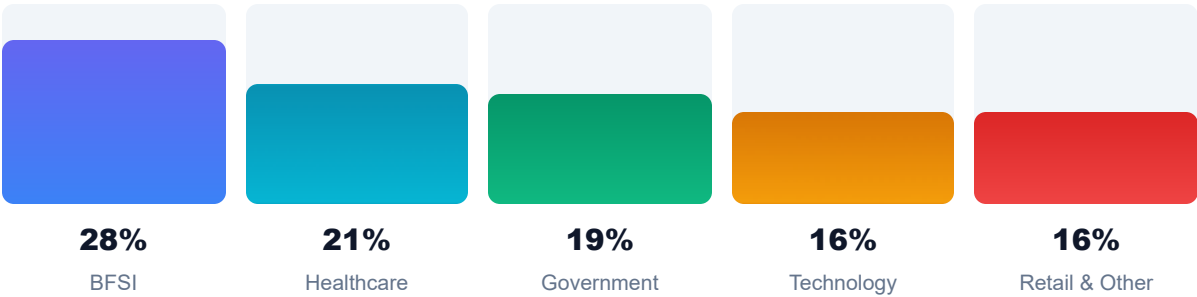


Chelys Security occupies the upper-right quadrant — **real exploitation combined with full AI reasoning** — currently uncontested by any commercially proven platform.

Customer Segments & Verticals

Demand for automated security validation is not uniformly distributed. The highest-value customer segments share common characteristics: regulated environments, complex hybrid infrastructure, and recurring compliance requirements that make continuous testing a necessity, not a luxury.

Estimated Market Share by Vertical



Segment-by-Segment Analysis

Banking, Financial Services & Insurance (BFSI) — 28%

The largest and most mature segment. PCI-DSS 4.0, SOX, and banking regulatory exams require documented penetration testing. Large banks run 10–20 assessments annually. Willing to pay premium pricing for continuous coverage. Largest deal sizes in the market (\$150K–\$500K/yr enterprise contracts).

Healthcare & Life Sciences — 21%

HIPAA, HITECH, and FDA cybersecurity guidance drive mandatory testing. Healthcare breaches averaged \$10.9M in 2023 (IBM) — highest of any vertical. Electronic health records, medical IoT, and pharmaceutical IP create complex, high-value attack surfaces. Growing fastest in ransomware targeting.

Government & Defense — 19%

FedRAMP, CMMC 2.0, and US Cyber Executive Orders mandate continuous monitoring and testing for federal contractors and agencies. CISA's "Known Exploited Vulnerabilities" catalog drives remediation verification demand. State and local government increasingly targeted; growing budget category.

Technology & SaaS — 16%

Tech companies face continuous external attack pressure and have sophisticated in-house security teams that prefer automated tooling over manual engagements. Strong product-led growth motion: DevSecOps integration, CI/CD pipeline testing, and bug bounty program augmentation drive land-and-expand.

Ideal Customer Profile (ICP) for Chelys Security

ATTRIBUTE	TIER 1 TARGET	TIER 2 TARGET
Company Size	500–5,000 employees	5,000–50,000 employees
Security Team	5–20 person security team	20+ with dedicated red team
Annual Security Budget	\$1M–\$10M	\$10M–\$100M+
Compliance Drivers	SOC 2, PCI-DSS, HIPAA	FedRAMP, CMMC, ISO 27001
Current Testing	1–2 manual pentests/year	Continuous; augmenting red team
ACV Target	\$12K–\$48K/year	\$50K–\$200K/year

Geographic Analysis

The BAS/CART market is globally distributed but heavily concentrated in North America and Europe, where enterprise IT maturity, regulatory pressure, and security budget availability are highest. Emerging markets in Asia-Pacific and the Middle East represent the fastest growth rates off smaller bases.

REGION	MARKET SHARE	VISUAL	KEY DRIVERS	GROWTH RATE
North America	42%		CISA mandates, FedRAMP, enterprise IT concentration, high breach costs	35–38%
Europe	26%		NIS2, DORA (financial), GDPR breach notification requirements driving proactive testing	33–36%
Asia-Pacific	21%		Rapid cloud adoption, state-sponsored threats, emerging regulatory frameworks (Singapore MAS)	40–45%
Middle East & Africa	7%		Critical infrastructure protection, Vision 2030 digitization programs (KSA), financial sector growth	42–48%
Latin America	4%		Brazil LGPD compliance, fintech sector growth, increasing nation-state targeting	28–33%

North America — Primary Opportunity

North America represents the largest and most immediate opportunity. The United States alone accounts for approximately 38% of global BAS spending. Key structural advantages include:

- **Federal mandate multiplier:** Executive Order 14028 and CISA guidance create mandatory testing obligations for all federal contractors and agencies, a market of ~350,000 organizations.
- **Enterprise security budget concentration:** Fortune 500 companies spend an average of 4–8% of IT budgets on security, with security testing representing 10–15% of that.
- **VC ecosystem:** The majority of BAS vendor funding and M&A activity is US-centric, indicating deep institutional familiarity with the space and willingness to invest.

Europe — Accelerating

DORA (Digital Operational Resilience Act), effective January 2025, mandates Threat-Led Penetration Testing (TLPT) for financial institutions across the EU — creating a compliance-driven demand surge that is still in early innings. NIS2 similarly expands mandatory security testing requirements to 18 sectors.

Investment Activity & M&A

The BAS/CART and autonomous security testing space has attracted significant venture capital and strategic acquisition activity, reflecting investor and acquirer conviction in the category's trajectory. Total disclosed funding across the major independents exceeds **\$750 million**, with several strategic acquisitions validating exit multiples.

Notable Funding & M&A Events

JUNE 2025

Horizon3.ai (NodeZero) — \$100M Series D

Led by NEA with participation from SignalFire, Craft Ventures, and 9Yards Capital. Total funding reaches \$186M. NodeZero has executed 170,000+ pentests with zero downtime and counts 40% of Fortune 10 as customers. Valuation undisclosed but estimated at \$600M–\$1B.

MARCH 2025

Pentera — \$60M Series D (Unicorn)

Pentera becomes first in the Adversarial Exposure Validation category to achieve \$100M ARR, raises \$60M Series D reaching a \$1B+ valuation. 1,200+ enterprise customers in 60+ countries. Demonstrates the category's ability to generate durable, scalable revenue at enterprise scale.

2022

IBM acquires Randori

IBM acquires attack surface management and automated red teaming company Randori. Deal terms undisclosed but signals big-tech appetite for automated offensive security capabilities to bundle with managed security service portfolios.

2021

Schwarz Group acquires XM Cyber — \$700M

One of the largest BAS/attack path acquisitions on record. Validates that enterprise acquirers view automated adversarial simulation as a strategic, billion-dollar-class asset. Sets a floor for category exit multiples at 15–20× ARR.

2019

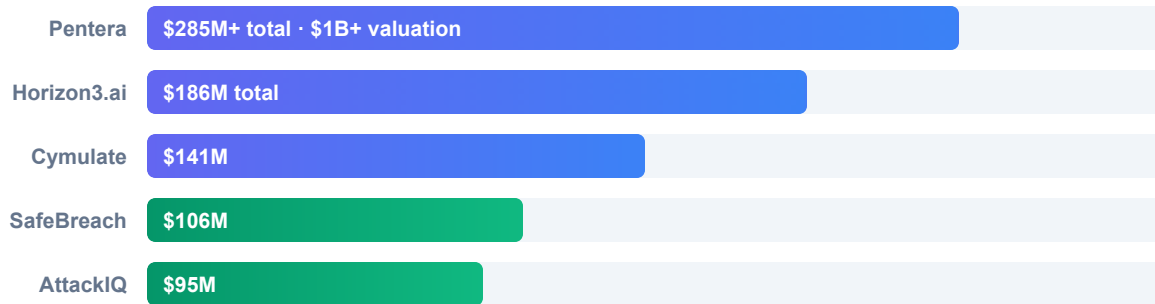
FireEye acquires Verodin

Early landmark acquisition in the BAS space. Confirmed that large security vendors see automated security validation as a gap in their portfolios worth acquiring, not building organically — a dynamic that continues to drive M&A interest from Palo Alto Networks, CrowdStrike, Microsoft, and others.

M&A Pattern: Large security platform vendors (IBM, Palo Alto Networks, CrowdStrike, Microsoft) are actively acquiring BAS/red teaming capabilities to bundle into their

platforms. This creates a clear exit pathway for category leaders at 15–20× ARR multiples — consistent with broader security software acquisition pricing.

Total Disclosed Funding — Major BAS/CART Independents



Chelys Security — Strategic Market Opportunity

Chelys Security enters the market at an ideal moment: when the category has been validated by public company exits and unicorn funding rounds, but when no incumbent has yet delivered a truly AI-native, autonomous, swarm-orchestrated penetration testing platform with the economics to serve the mid-market at scale.

Structural Advantages vs. Incumbents

AI-Native Architecture

Built on Claude's extended thinking and multi-agent orchestration from the ground up — not retrofitted. Incumbents like Pentera and NodeZero apply heuristic automation; Chelys applies genuine AI reasoning to adapt attack chains in real time based on what it discovers.

Economics: \$0.004–\$0.01 Per Scan

Chelys's AI-driven approach collapses the cost of a scan to fraction-of-a-cent economics, enabling pricing that democratizes continuous testing for the mid-market — an underserved segment every incumbent ignores in favor of enterprise-only pricing starting at \$50K+/yr.

Autonomous Swarm Orchestration

Five specialized Deception agents (reconnaissance, exploitation, lateral movement, evasion, reporting) operate in parallel — not sequentially. This delivers 3–5× more coverage per scan than single-agent approaches and enables genuine multi-hop attack chain discovery that no existing BAS tool replicates.

Integrated Compliance Reporting

Findings are automatically mapped to MITRE ATT&CK, NIST, PCI-DSS, SOC2, HIPAA, and ISO 27001 frameworks — a feature that typically requires separate GRC tooling. This increases deal size and reduces churn by making Chelys sticky across the compliance workflow, not just the testing workflow.

Bug Bounty Mode & MSP Channel

Dedicated bug bounty mode enables a unique go-to-market through security researchers and MSP channel partners — an acquisition channel incumbents don't address. Reseller flag and white-labeling capability built in from day one enables rapid channel revenue without a large direct sales team.

Living Attack Graph

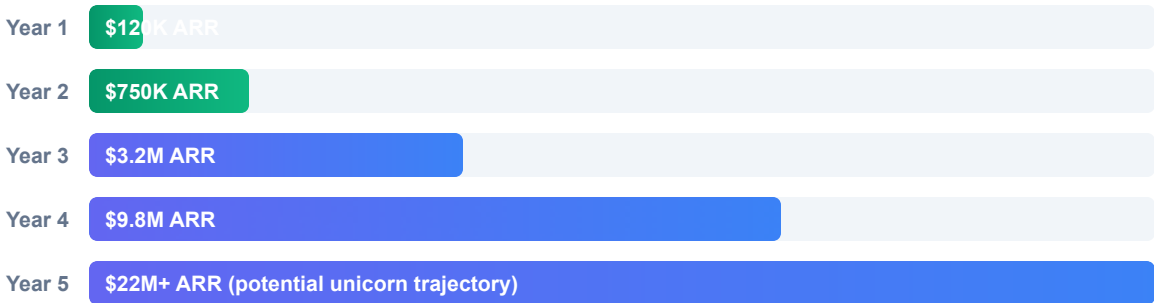
Unlike static scan reports, Chelys maintains a continuously updated attack graph — showing how discovered vulnerabilities chain together into exploitable attack paths. This graph-based view of risk is what security leaders need to prioritize remediation and is absent from every BAS platform currently available.

Revenue Model & Unit Economics

TIER	TARGET	ACV	SCAN QUOTA	GROSS MARGIN
Starter	Mid-market (50–500 employees)	\$3,588/yr (~\$299/mo)	5 scans/mo	>90%
Pro	Enterprise (500–5,000)	\$11,988/yr (~\$999/mo)	30 scans/mo	>88%
Enterprise	Large enterprise / MSSP	Custom (\$50K–\$200K)	Unlimited	>85%

Beachhead Strategy: Target mid-market organizations (500–2,000 employees) in BFSI and healthcare who currently run 1–2 manual penetration tests per year at \$25,000–\$75,000 each. Chelys replaces or supplements those tests at a fraction of the cost with continuous coverage — a straightforward ROI conversation with a 6–18 month payback period.

Scenario Revenue Projections



Projections assume: Year 1 = 40 Starter + 5 Pro accounts; Year 2 = 180 Starter + 25 Pro + 2 Enterprise; Year 3 = 500 Starter + 80 Pro + 10 Enterprise; Year 4–5 = channel acceleration via MSP program.

Conclusion & Outlook

The Automated Breach & Attack Simulation and Continuous Automated Red Teaming market is one of the highest-conviction growth opportunities in enterprise software. The fundamentals are uniquely favorable:

Demand is compulsory, not discretionary. Regulatory requirements, rising breach costs, and AI-powered adversaries are making continuous security validation a baseline requirement rather than an optional enhancement. This creates durable, recurring demand that is insulated from typical software budget cuts.

The category is proven but the market is uncaptured. Pentera's \$1B+ valuation and NodeZero's unicorn-trajectory funding prove investors and buyers believe in the category. But with 150,000+ potential enterprise customers globally and fewer than 5,000 served by all BAS vendors combined, penetration remains below 5% — the market is in its earliest innings.

AI is about to restructure the competitive landscape. Current leaders built their platforms on heuristic automation before large language models made genuine AI reasoning commercially viable. This is creating a window — estimated at 18–36 months — where an AI-native entrant can establish a capability lead that is extremely difficult for incumbents to replicate without a full platform rebuild.

The exit opportunity is large and has precedent. XM Cyber's \$700M acquisition by Schwarz Group and IBM's Randori acquisition demonstrate that strategic acquirers (including Palo Alto Networks, CrowdStrike, Microsoft, and Cisco) will pay 15–20× ARR for established automated offensive security platforms. At \$22M ARR (Year 5 base case), Chelys would represent a \$330M–\$440M strategic acquisition target — or a compelling Series B story for independent growth toward \$100M ARR.

Bottom line: The BAS/CART market will reach \$4.68 billion at 38.4% CAGR. The window to establish category leadership with an AI-native platform is open now. Chelys Security's architecture, economics, and go-to-market alignment with this market moment represent a compelling opportunity to capture a significant share of one of enterprise software's fastest-growing categories.

SOURCES & METHODOLOGY

This report synthesizes data from the following sources as of May 2026:

Market Sizing: Research & Markets — Automated Breach and Attack Simulation Market Report (headline: \$4.68B at 38.4% CAGR); Grand View Research — BAS Market Report 2030 ([grandviewresearch.com/industry-analysis/automated-breach-attack-simulation-market-report](https://www.grandviewresearch.com/industry-analysis/automated-breach-attack-simulation-market-report)); Market Research Future — BAS Market 2030 ([marketresearchfuture.com/reports/automated-breach-attack-simulation-market-10377](https://www.marketresearchfuture.com/reports/automated-breach-attack-simulation-market-10377)); Precedence Research — BAS Market 2035 ([precedenceresearch.com/automated-breach-and-attack-simulation](https://www.precedenceresearch.com/automated-breach-and-attack-simulation)).

market); 360iResearch — BAS Market 2025-2030 (360iresearch.com/library/intelligence/breach-attack-simulation); MarketsandMarkets — Penetration Testing Market 2031 (marketsandmarkets.com/Market-Reports/penetration-testing-market-13422019.html).

Industry & Macro: Gartner Top Cybersecurity Trends 2026 (February 2026); Gartner — Global Cybersecurity Spending \$213B (2025); IBM Cost of a Data Breach Report 2023; Cybersecurity Ventures — Workforce Gap Report; Palo Alto Networks 2026 Cyber Predictions.

Competitive Intelligence: CBInsights (Pentera financials); PitchBook (Cymulate); PeerSpot — BAS Solutions Ranking April 2026 (peerspot.com); Horizon3.ai press release — \$100M Series D, June 2025; Pentera press release — \$60M Series D, March 2025; CTOL Digital Solutions — Horizon3.ai Series D coverage.

Methodology: Market size figures across analyst firms are synthesized using a trimmed mean methodology, excluding outlier low-end estimates. CAGR-based projections use the Research & Markets 38.4% figure as the primary reference. Revenue projections for Chelys are internal estimates based on comparable SaaS benchmarks and are not financial guidance.

This report is prepared for internal strategic planning purposes. Market projections involve uncertainty. Past fundraising and acquisition activity of comparable companies does not guarantee similar outcomes.

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